

香港中文大學
The Chinese University of Hong Kong

Formal Languages and Automata Theory

Lecture 1

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Outline



Mathematical Preliminaries and Notation



Three Basic Concepts



Some Applications (omit)

Mathematical Preliminaries

- Sets
- Functions
- Relations
- Graphs
- Proof Techniques

SETS

A set is a collection of elements

$$A = \{1, 2, 3\}$$

$$B = \{train, bus, bicycle, airplane\}$$

We write

$1 \in A$ ⇨ 1 is an element of the set A

$ship \notin B$ ⇨ ship is not an element of the set B

Set Representations

$C = \{ a, b, c, d, e, f, g, h, i, j, k \}$

$C = \{ a, b, \ldots, k \}$ ⟶ *finite set*

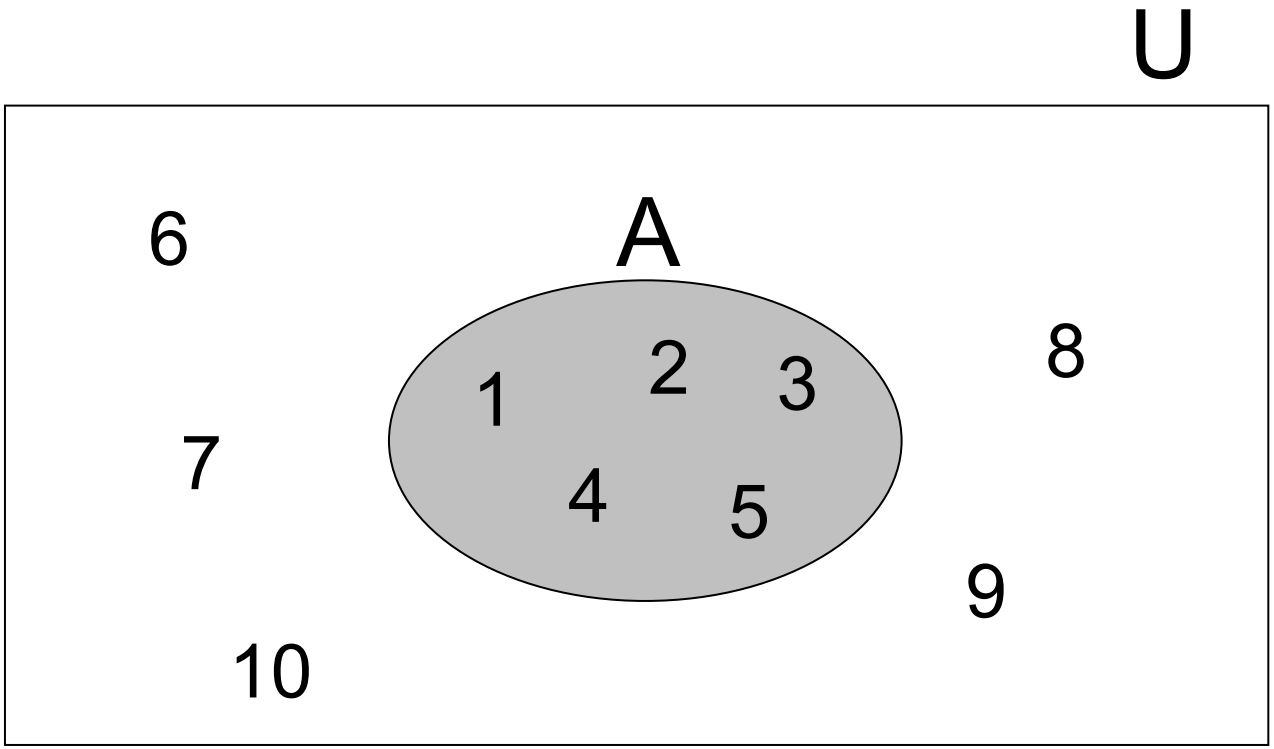
$S = \{ 2, 4, 6, \ldots \}$ ⟶ *infinite set*

$S = \{ j : j > 0, \text{ and } j = 2k \text{ for some } k>0 \}$

$S = \{ j : j \text{ is nonnegative and even} \}$

} Explicit notation

$A = \{ 1, 2, 3, 4, 5 \}$



Universal Set: all possible elements

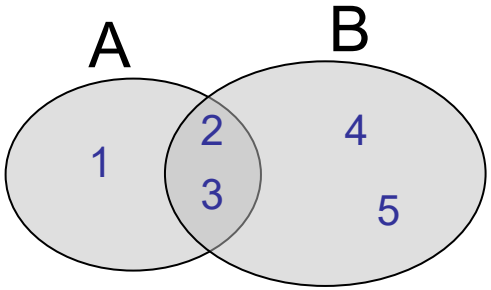
$U = \{ 1 , \ldots , 10 \}$

Set Operations

A = { 1, 2, 3 } B = { 2, 3, 4, 5 }

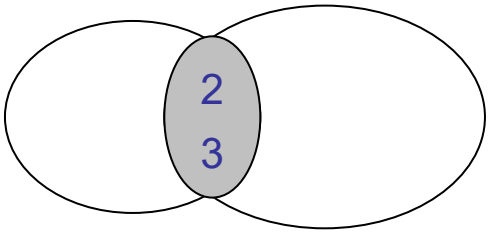
- Union (U)

  A U B = { 1, 2, 3, 4, 5 }



- Intersection (∩)

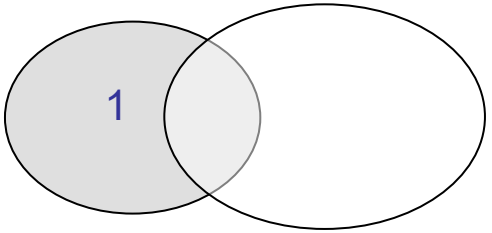
  A ∩ B = { 2, 3 }



- Difference (-)

  A - B = { 1 }

  B - A = { 4, 5 }

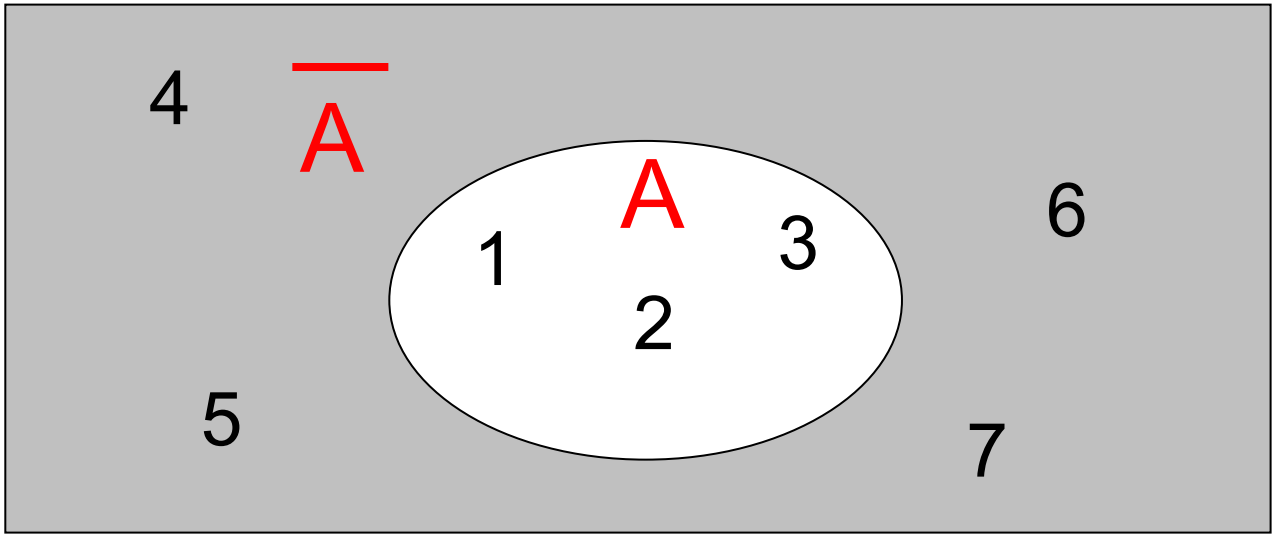


Venn diagrams

- Complement

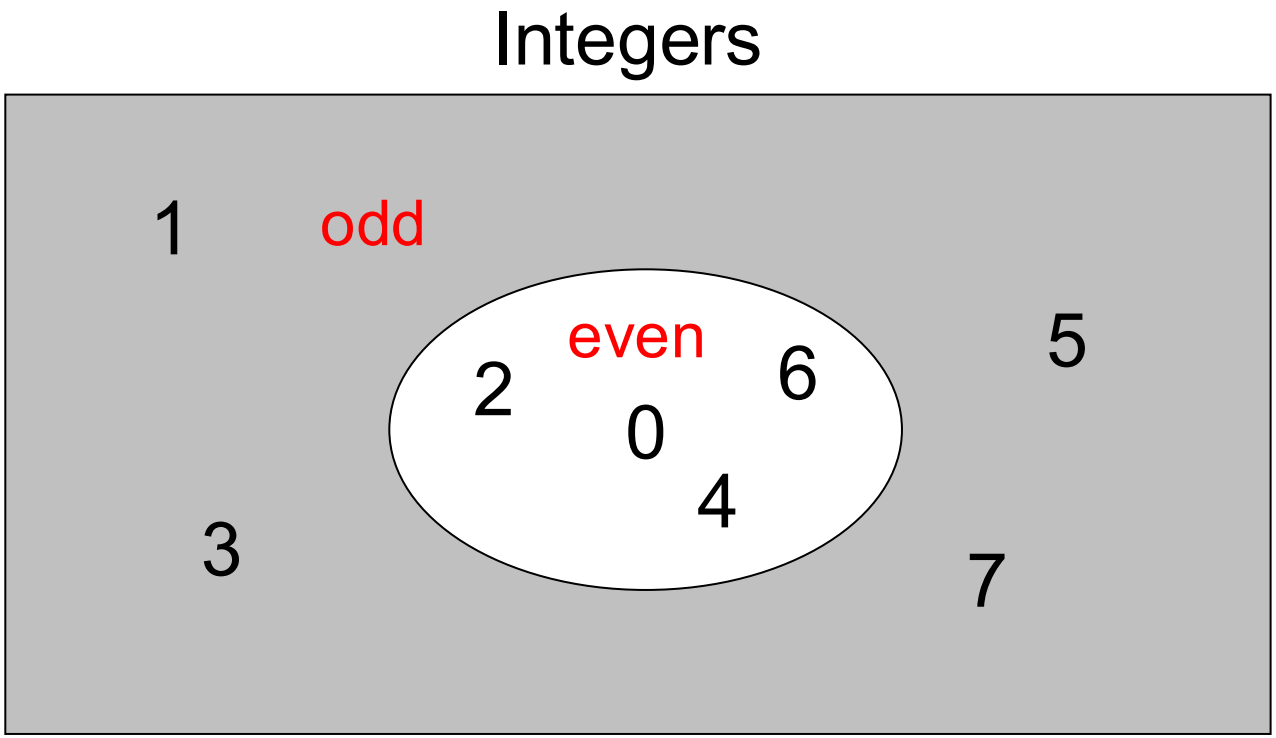
Universal set = {1, …, 7}

$A = \{ 1, 2, 3 \} \Rightarrow \overline{A} = \{ 4, 5, 6, 7 \}$



$\overline{\overline{A}} = A$

$\overline{\{ \text{even integers} \}} = \{ \text{odd integers} \}$



DeMorgan's Laws

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

Empty, Null Set: $\emptyset$

$\emptyset = \{\ \}$ ⇨ The set with no elements

$S \cup \emptyset = S$

$S \cap \emptyset = \emptyset$

$S - \emptyset = S$

$\emptyset - S = \emptyset$

$\overline{\emptyset}$ = Universal Set

Subset

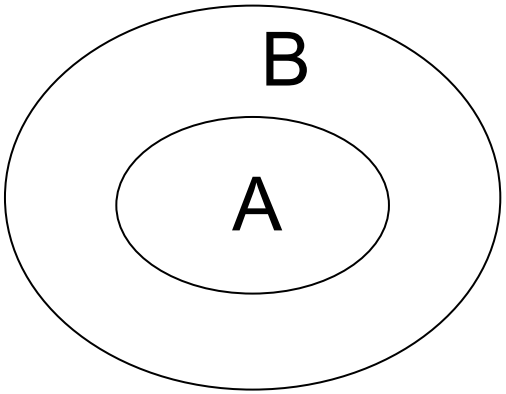
$A = \{ 1, 2, 3 \}$ $B = \{ 1, 2, 3, 4, 5 \}$

$A \subseteq B$

Subset: If every element of A is also an element of B

Proper Subset: If $A \subseteq B$, but B contains an element not in A

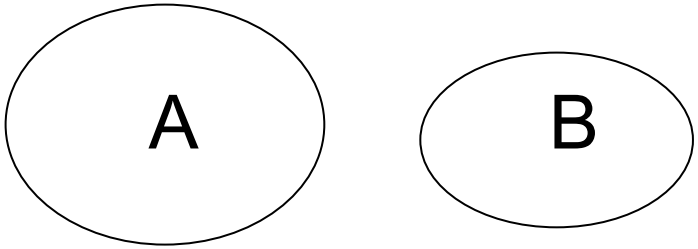
$A \subset B$



Disjoint Sets

$A = \{ 1, 2, 3 \}$ $B = \{ 5, 6\}$

$A \cap B = \emptyset$



Set Cardinality

- For finite sets

  $A = \{ 2, 5, 7 \}$

  $|A| = 3$ (set size)

Powersets

A powerset is a set of sets

$S = \{ a, b, c \}$

Powerset of S = the set of all the subsets of S

Example 1.1

$2^S = \{ \emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\} \}$

Observation: if S is finite, then $|2^S| = 2^{|S|}$ ($8 = 2^3$)

Cartesian Product

Example 1.2

$A = \{ 2, 4 \}$ $B = \{ 2, 3, 5, 6 \}$

$$A \times B = \{ (2, 2), (2, 3), (2, 5), (2, 6), (4, 2), (4, 3), (4, 5), (4, 6) \}$$

$$|A \times B| = |A| \, |B|$$

Note that the **order** in which the elements of a pair are written matters

The pair (4, 2) is in A X B, but (2, 4) is not

Partition

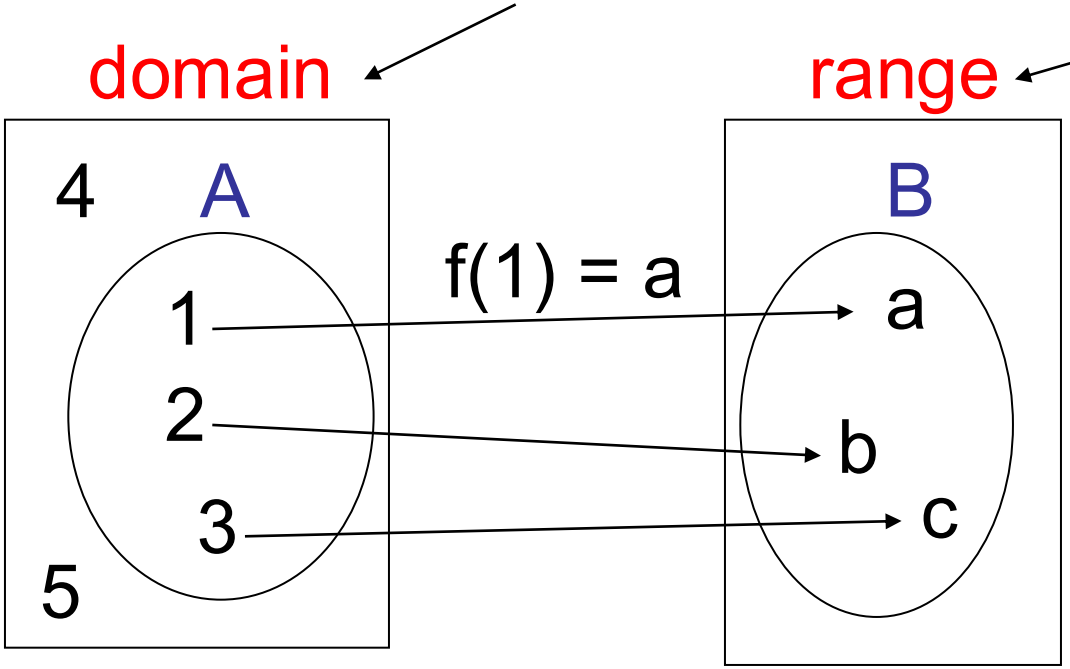
A set can be divided by separating it into a number of subsets. Suppose that S_1, S_2, …, S_n are subsets of a given set S and that the following holds:

1. The subset S_1, S_2, …, S_n are mutually disjoint;
2. $S_1 \cup S_2 \cup \ldots \cup S_n = S$;
3. None of the S_i is empty.

Then S_1, S_2, …, S_n is called a ***partition*** of S.

FUNCTIONS

Rules that assign to elements of one set a unique element of another set



$f : A \rightarrow B$

If A = domain

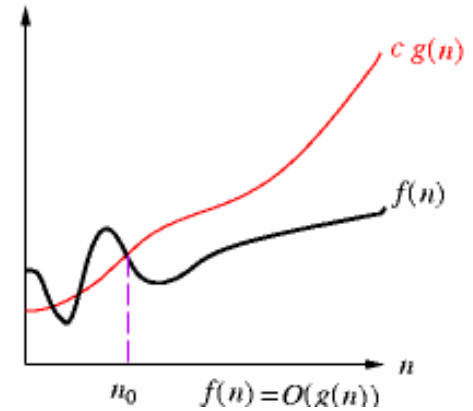
then f is a total function

otherwise f is a partial function

Asymptotic Analysis

O: Upper Bounding Function

- **Def:** $f(n) = O(g(n))$ if $\exists\ c > 0$ and $n_0 > 0$ such that $0 \le \mathbf{f(n)} \le \mathbf{cg(n)}$ for all $n \ge n_0$.
- Intuition: $f(n)$ "$\le$" $g(n)$ when we ignore constant multiples and small values of n.
- How to show *O* (Big-Oh) relationships?
  - $f(n) = O(g(n))$ implies that $\lim_{n \to \infty} \frac{f(n)}{g(n)} = c$ for some $c \ge 0$, if the limit exists.



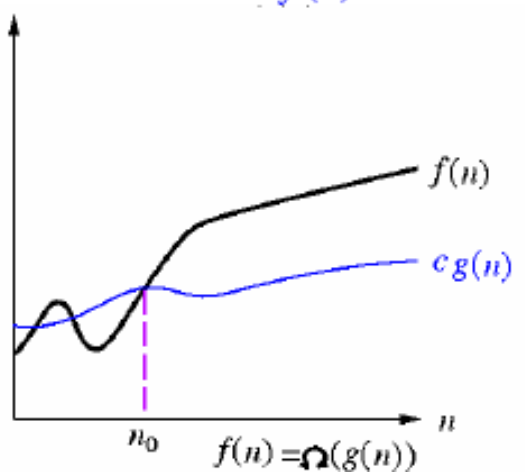
1. $3n^2 + n = O(n^2)$?
2. $3n^2 + n = O(n)$?
3. $3n^2 + n = O(n^3)$?

Asymptotic Analysis

Ω : Lower Bounding Function

- **Def:** $f(n) = \Omega(g(n))$ if $\exists\ c > 0$ and $n_0 > 0$ such that $0 \le cg(n) \le f(n)$ for all $n \ge n_0$.
- Intuition: $f(n)$ " $\ge$ " $g(n)$ when we ignore constant multiples and small values of n.
- How to show Ω (Big-Omega) relationships?
  - $f(n) = \Omega(g(n))$ implies that $\lim_{n \to \infty} \frac{g(n)}{f(n)} = c$ for some $c \ge 0$ if the limit exists.

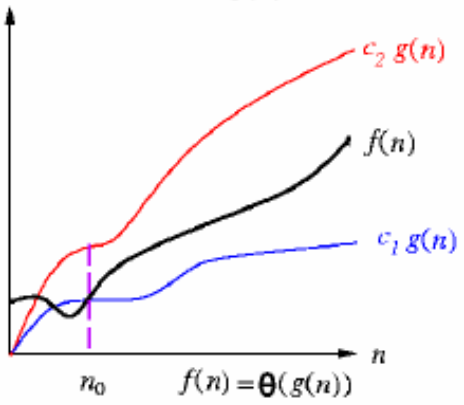
1. $3n^2 + n = \Omega(n^2)$?
2. $3n^2 + n = \Omega(n)$?
3. $3n^2 + n = \Omega(n^3)$?



Asymptotic Analysis

θ: Tightly Bounding Function

- **Def:** $f(n)$= $\theta(g(n))$ if $\exists$ c_1, c_2 >0 and $n_0 > 0$ such that $0 \le c_1 g(n) \le f(n) \le c_2\, g(n)$ for all $n \ge n_0$.
- Intuition: $f(n)$ " = " $g(n)$ when we ignore constant multiples and small values of n.
- How to show θ relationships?
  - Show both "big Oh" (O) and "Big Omega" (Ω) relationships.
  - $f(n) = \theta(g(n))$ implies that $\lim_{n \to \infty} \frac{f(n)}{g(n)} = c$ for some $c > 0$, if the limit exists.



1. $3n^2 + n = \theta(n^2)$?
2. $3n^2 + n = \theta(n)$?
3. $3n^2 + n = \theta(n^3)$?

Example 1.3

f(n) = $2n^2 + 3n$,
g(n) = n^3,
h(n) = $10n^2 + 100$.

f(n) = (g(n)),
g(n) = (h(n))
f(n) = (h(n))

RELATIONS

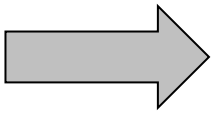
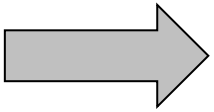
Relations are more general than functions:
In a function, each element of the domains has **exactly one** associated element in the range;
In a relation, there may be **several** such elements in the range.

$$R = \{(x_1, y_1), (x_2, y_2), (x_3, y_3), \ldots\}$$

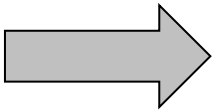
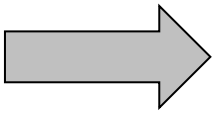
$$x_i \, R \, y_i$$

e. g. if R = '>': $2 > 1$, $3 > 2$, $3 > 1$

Equivalence Relations (≡)

- Reflexive: x R x
- Symmetric: x R y  y R x
- Transitive: x R y and y R z  x R z

Example: R ≡ '='

- x = x
- x = y  y = x
- x = y and y = z  x = z

Example 1.4

On the set of nonnegative integers, we can define a relation

$$x \equiv y$$

If and only if

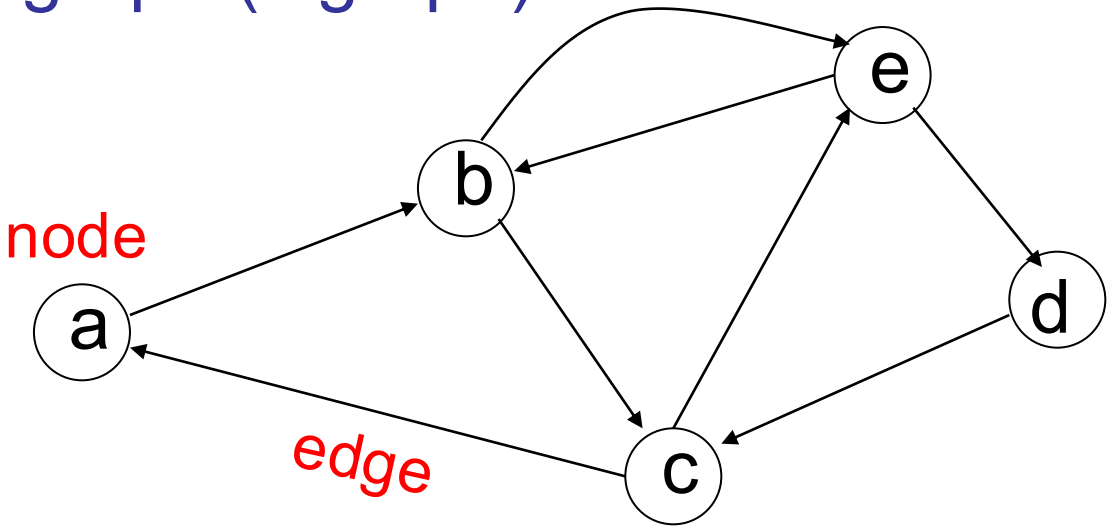
$$x \bmod 3 = y \bmod 3.$$

Then $2 \equiv 5$, $12 \equiv 0$, and $0 \equiv 36$.

Clearly this is an equivalence relation, as it satisfies reflexivity, symmetry, and transitivity.

GRAPHS

A directed graph (digraph)



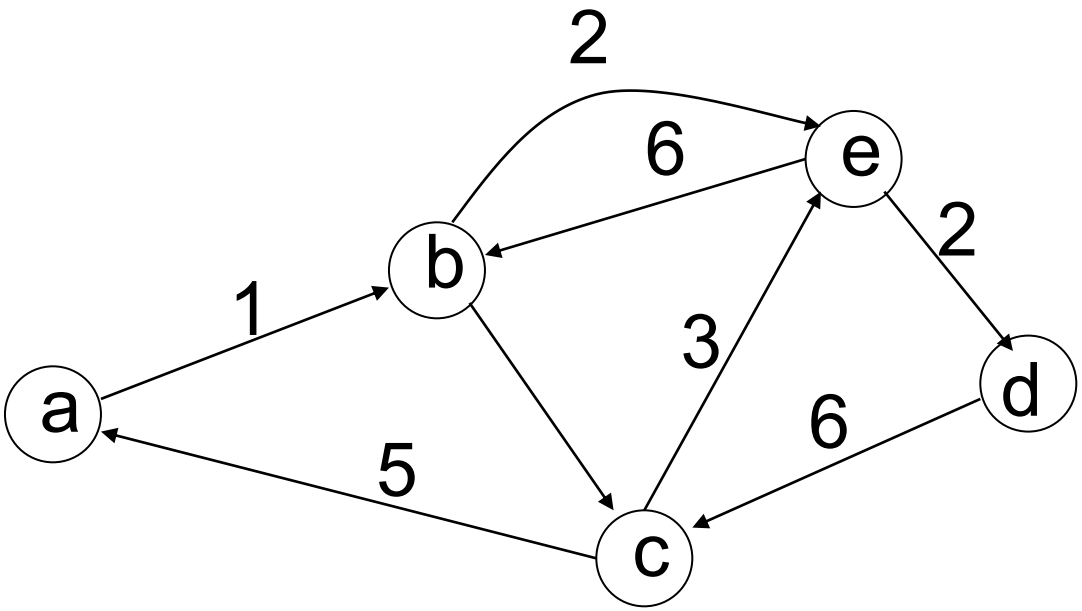
- Nodes (Vertices)

$$V = \{ a, b, c, d, e \}$$

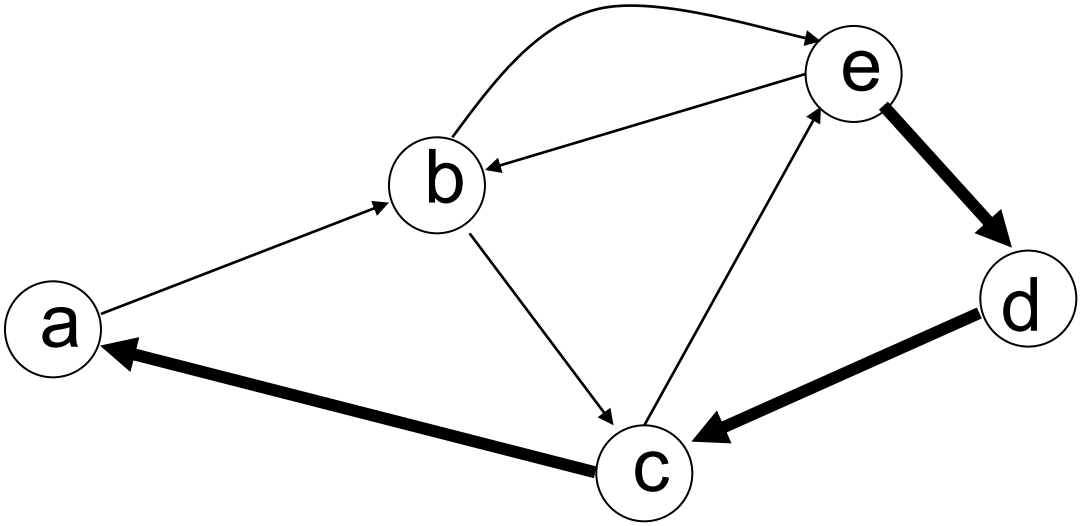
- Edges

$$E = \{ (a,b), (b,c), (b,e), (c,a), (c,e), (d,c), (e,b), (e,d) \}$$

Labeled Graph



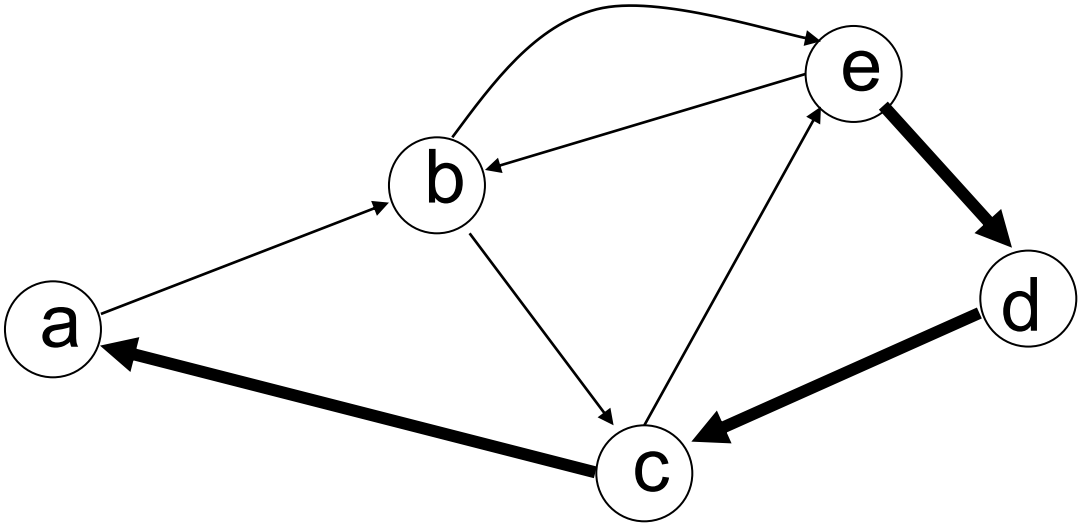
Walk



Walk is a sequence of adjacent edges

(e, d), (d, c), (c, a)

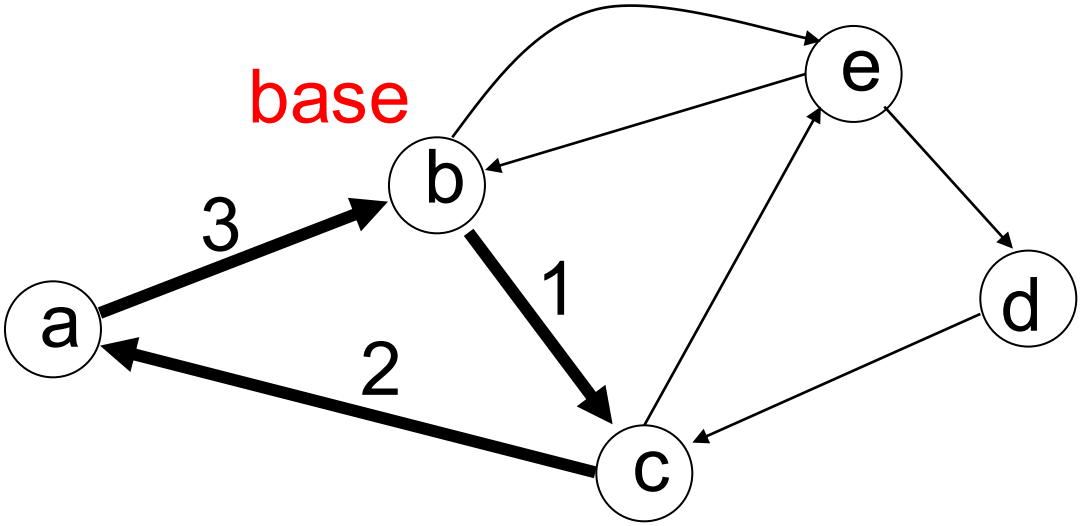
Path



Path: a walk where no edge is repeated

Simple path: no node is repeated

Cycle

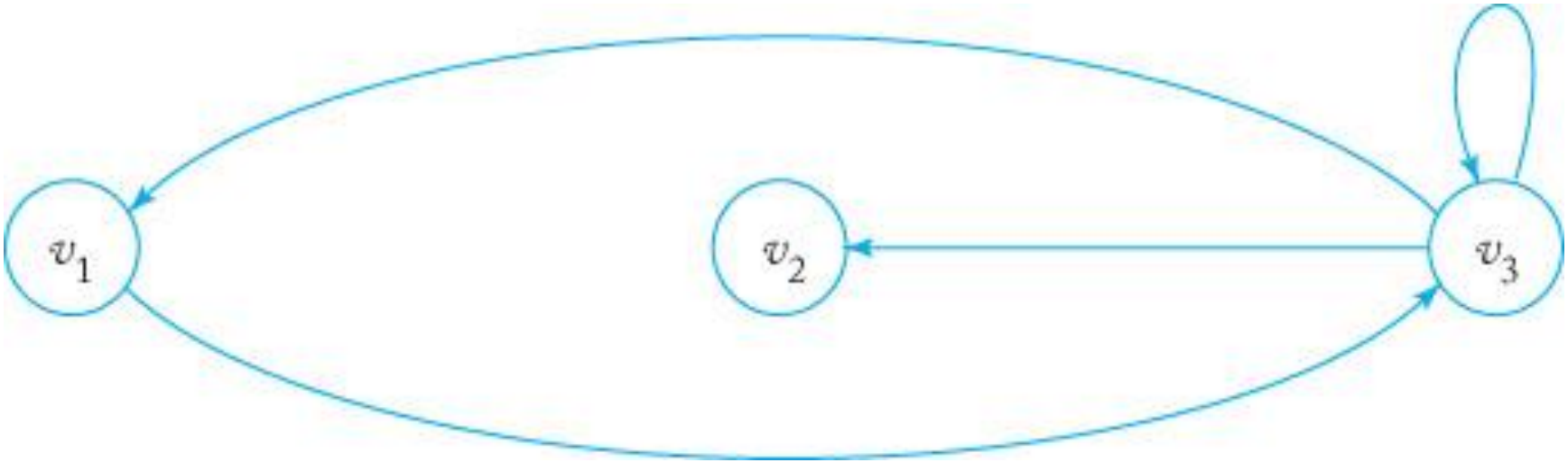


Cycle: a walk from a node (base) to itself

Simple cycle: only the base node is repeated

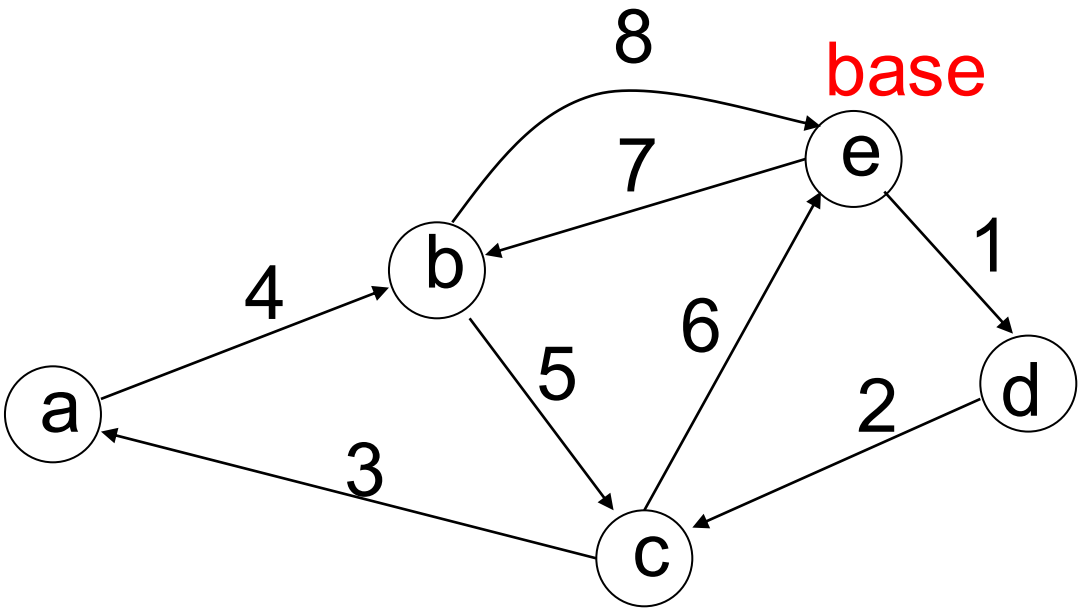
Loop

An edge from a vertex to itself



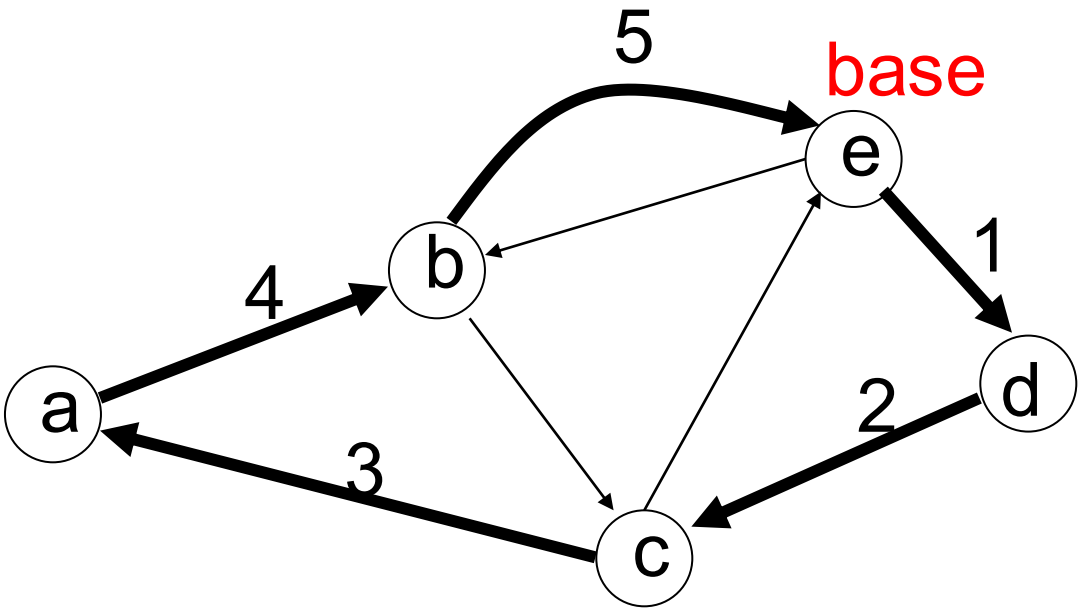
- $(v_1, v_3), (v_3, v_2)$ is a simple path from v_1 to v_2
- $(v_1, v_3), (v_3, v_3), (v_3, v_1)$ is a cycle (not simple one)
- There is a loop on vertex v_3

Euler Tour



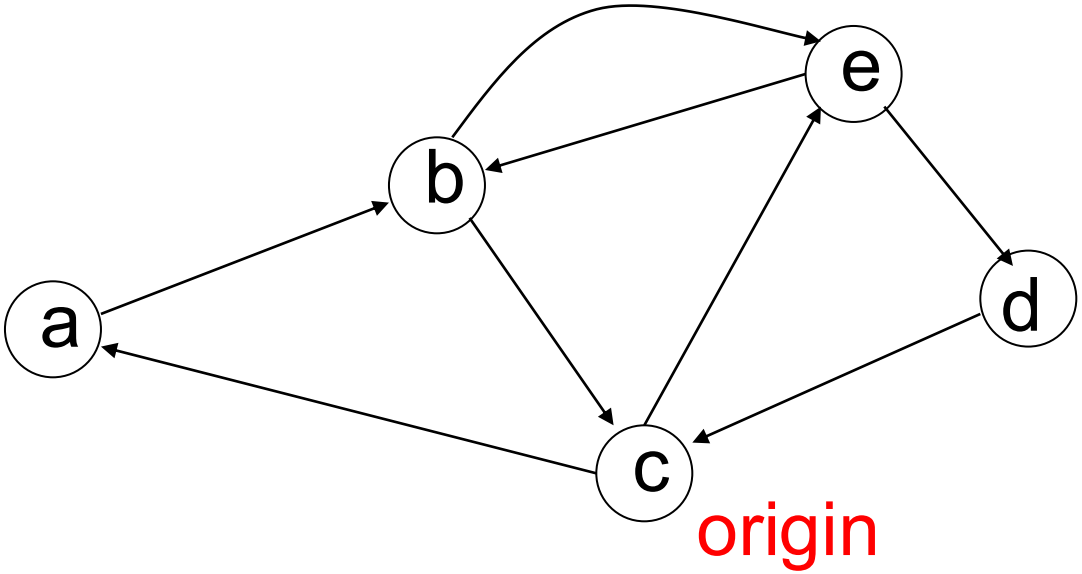
A cycle that contains each edge once

Hamiltonian Cycle

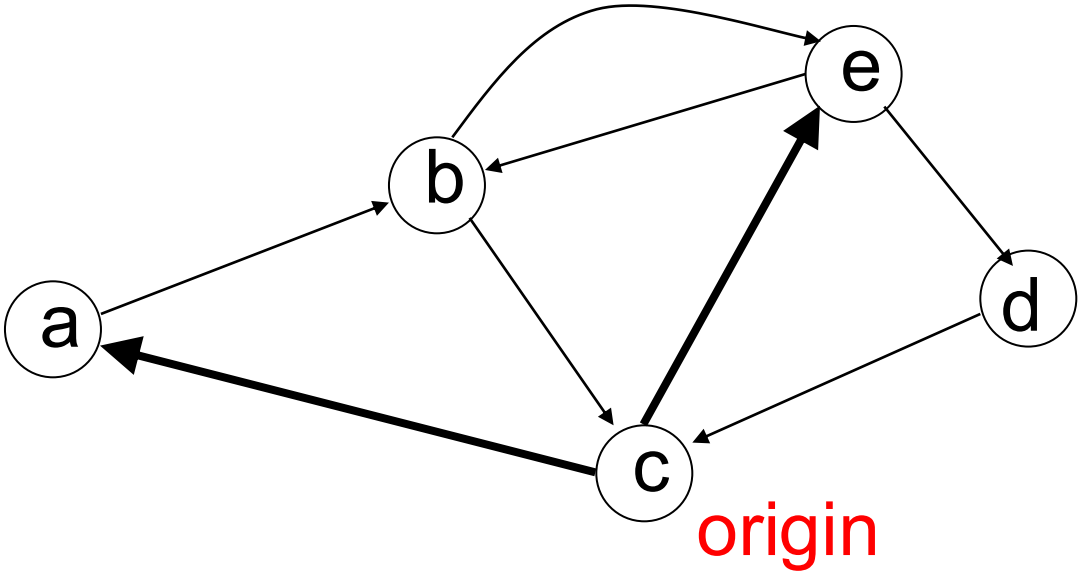


A simple cycle that contains all nodes

Finding All Simple Paths



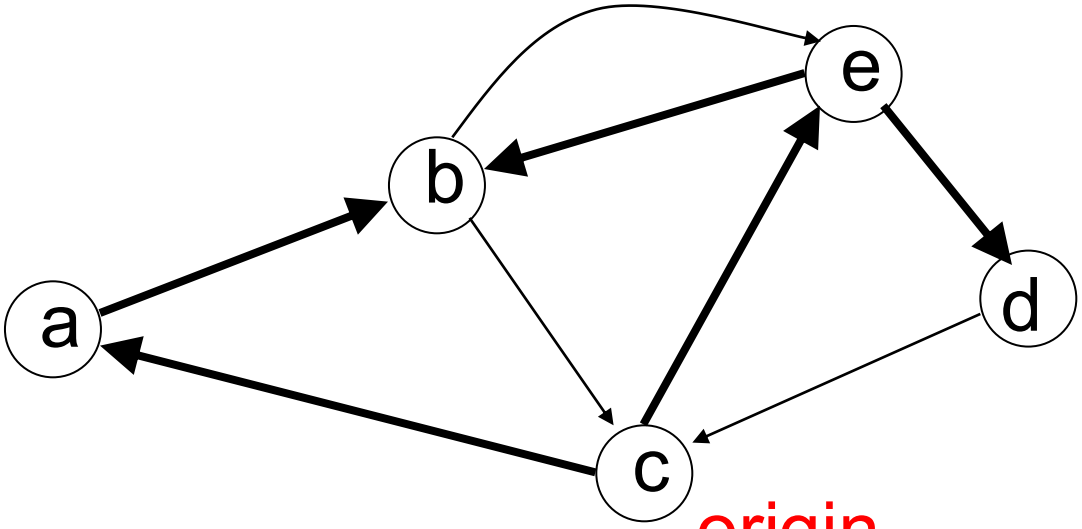
Step 1



(c, a)

(c, e)

Step 2



(c, a)

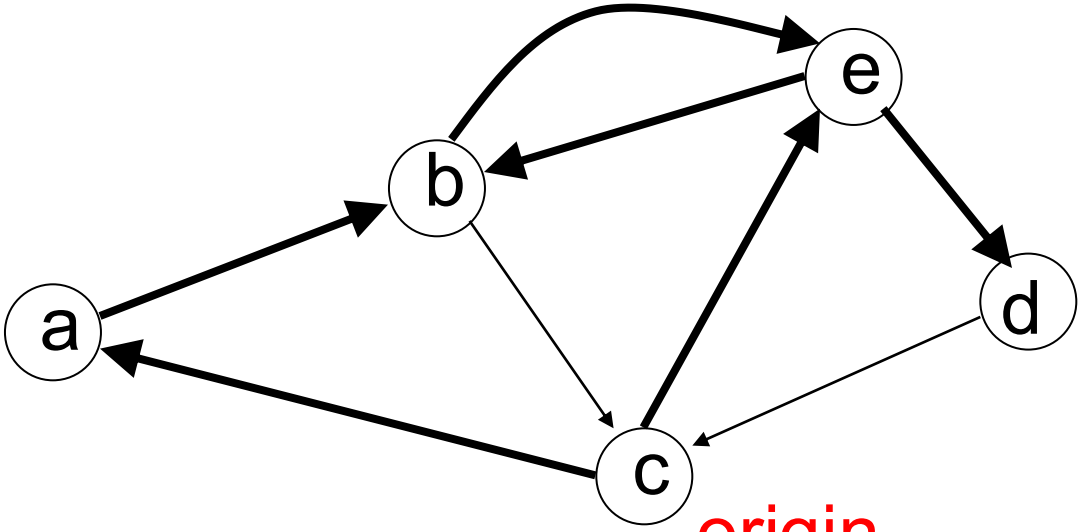
(c, a), (a, b)

(c, e)

(c, e), (e, b)

(c, e), (e, d)

Step 3



(c, a)

(c, a), (a, b)

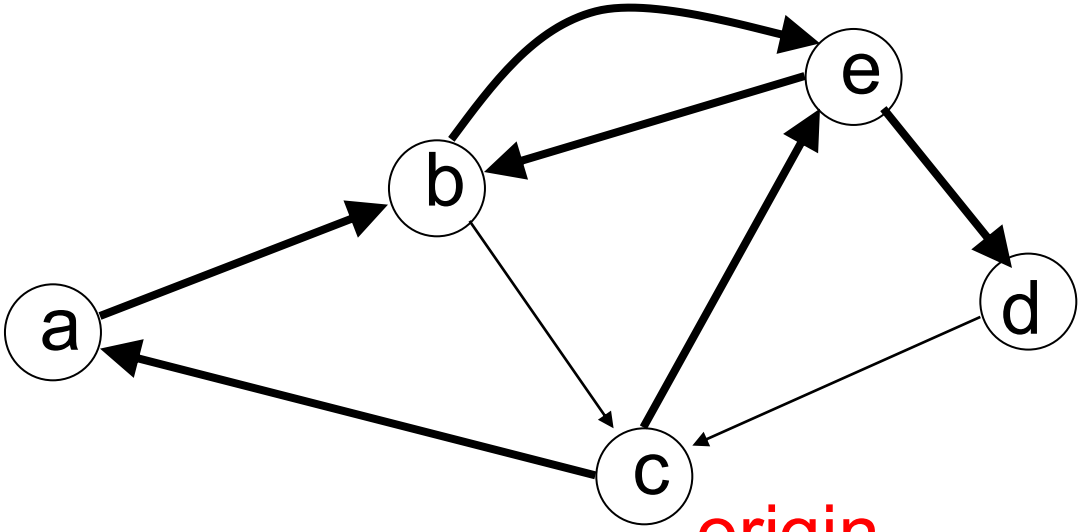
(c, a), (a, b), (b, e)

(c, e)

(c, e), (e, b)

(c, e), (e, d)

Step 4



(c, a)

(c, a), (a, b)

(c, a), (a, b), (b, e)

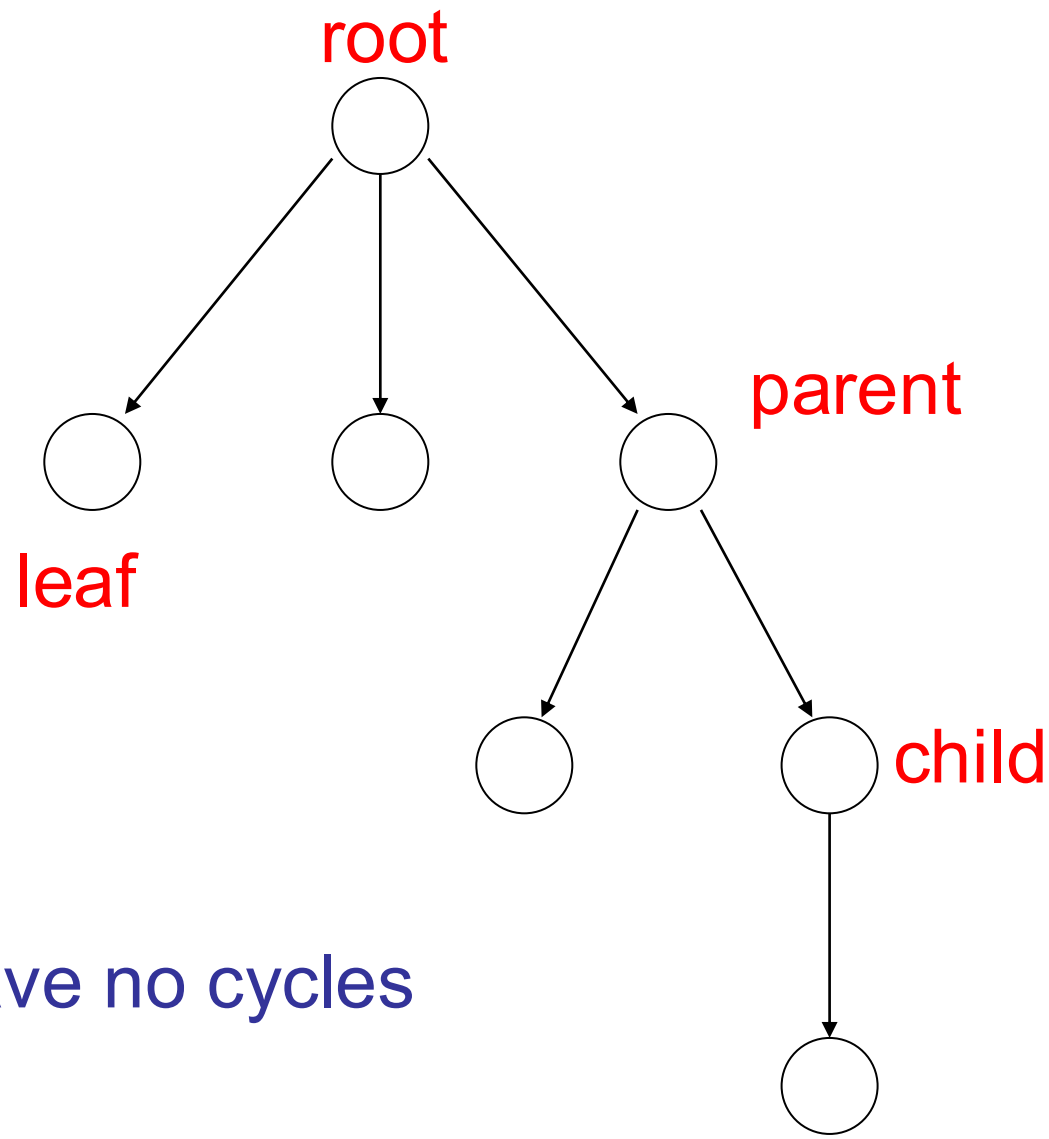
(c, a), (a, b), (b, e), (e,d)

(c, e)

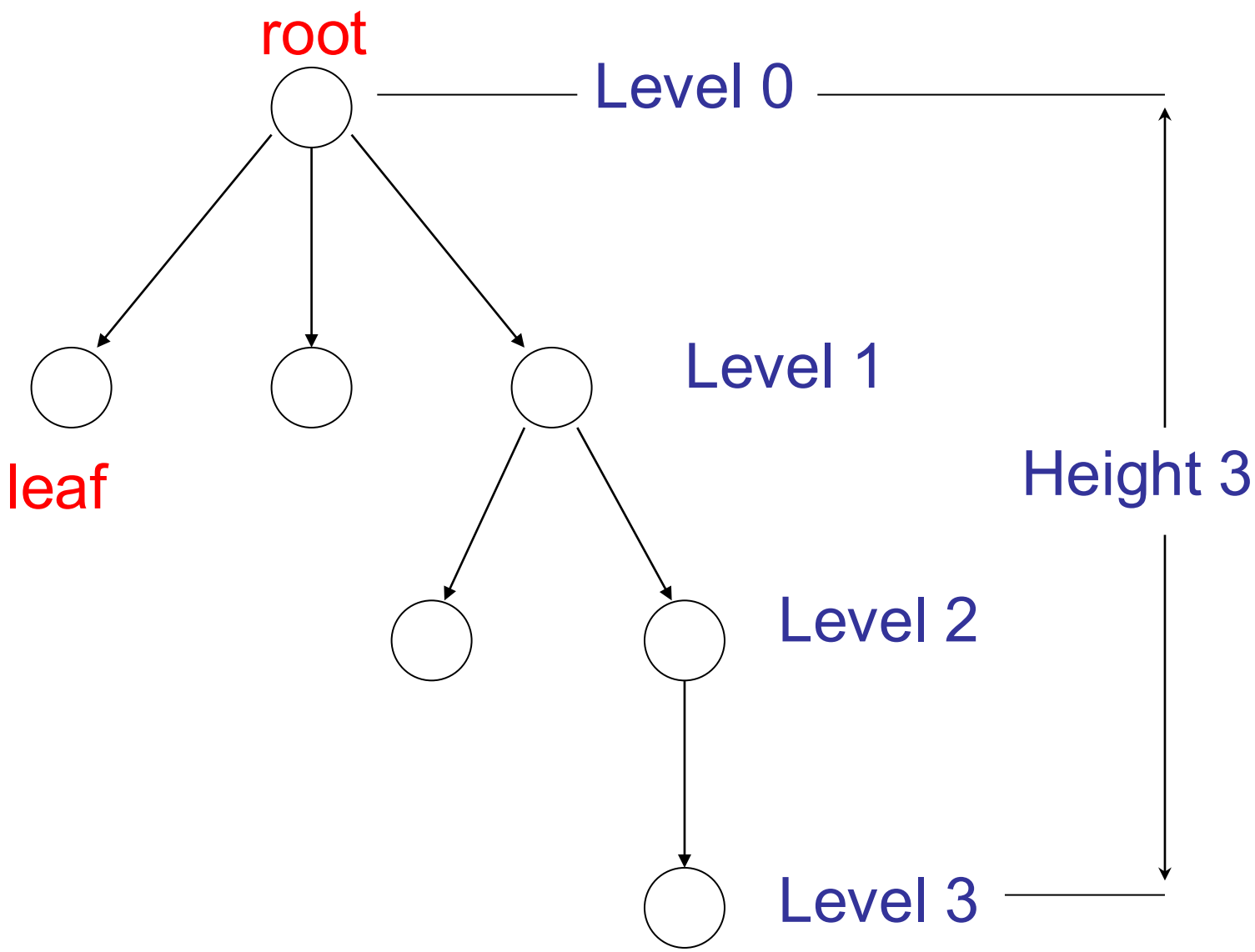
(c, e), (e, b)

(c, e), (e, d)

Trees



Trees have no cycles


root
Level 0
Level 1
leaf
Height 3
Level 2
Level 3

Proof Techniques

- Direct/Constructive Proof
- Proof by Induction
- Proof by Contradiction

Direct/Constructive Proof

- If X, then Y
- Assume X is true, show directly that Y is true.
  (e.g. X = it rains, Y = sidewalk will wet)
  - Example:
    - For integers *a,b*: If *a* and *b* are odd, then ab is odd.

Direct/Constructive Proof

- Perform the multiplication directly
  - $ab = (2x + 1)(2y + 1)$
    $= 4xy + 2x + 2y + 1$
    $= 2(2xy + x + y) + 1$

  So $z = 2xy + x + y$

Not only did you prove that a z exists, you constructed an "algorithm" for generating this z.

This is an example of a constructive proof.

Induction

We have statements $P_1, P_2, P_3, \ldots$

If we know

- for some b that $P_1, P_2, \ldots, P_b$ are true
- for any k >= b that

$$P_1, P_2, \ldots, P_k \text{ imply } P_{k+1}$$

Then

Every P_i is true

Proof by Induction

- Inductive basis

  Find P_1, P_2, …, P_b which are true

- Inductive hypothesis

  Let's assume P_1, P_2, …, P_k are true,

  for any k >= b

- Inductive step

  Show that P_{k+1} is true

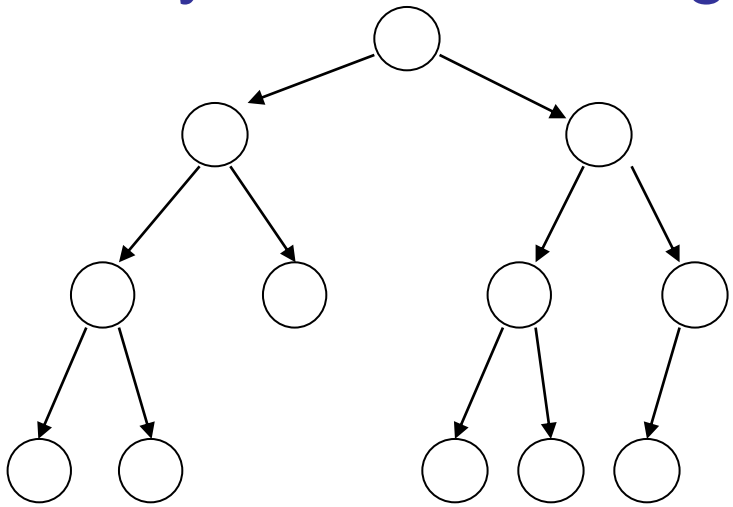
Example 1.5

Theorem: A binary tree of height n has at most 2^n leaves.

Proof by induction:

let L(i) be the maximum number of

leaves of any subtree at height i



We want to show: $L(i) <= 2^i$

- Inductive basis

  $L(0) = 1$ (the root node) ◯

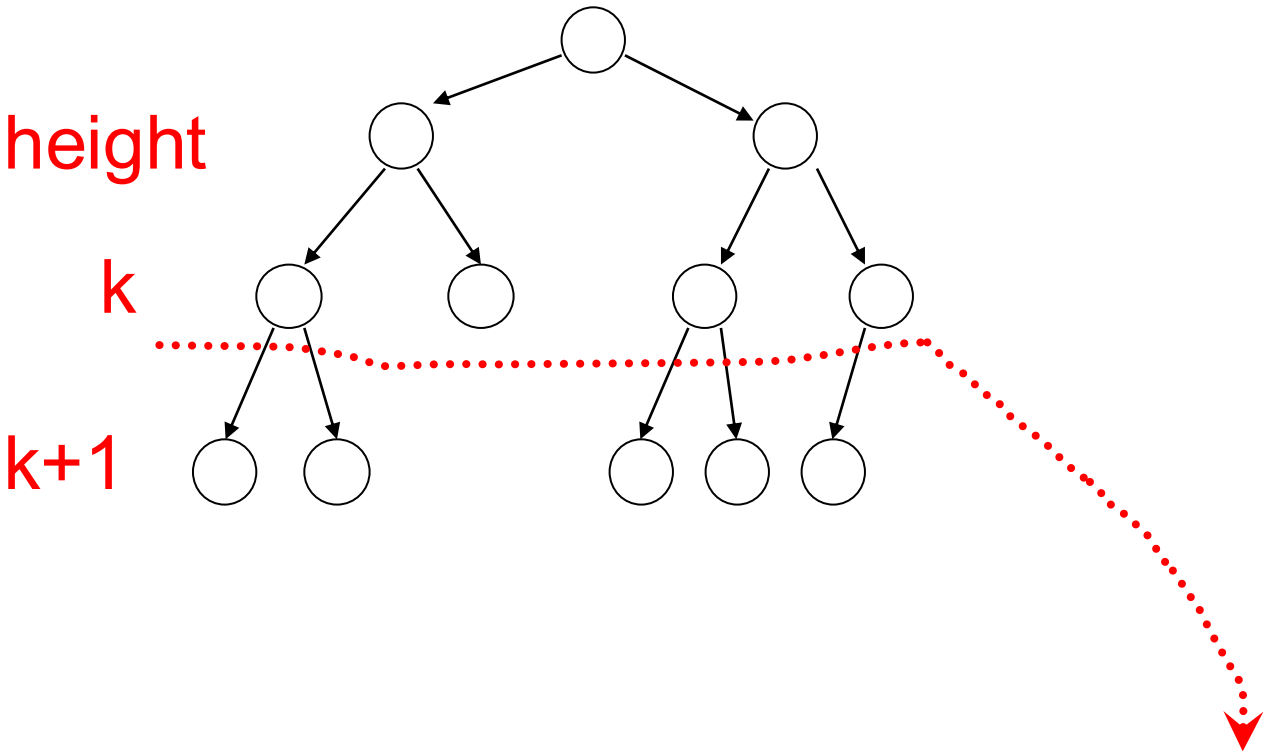
- Inductive hypothesis

  Let's assume $L(i) <= 2^i$ for all $i = 0, 1, \ldots, k$

- Induction step

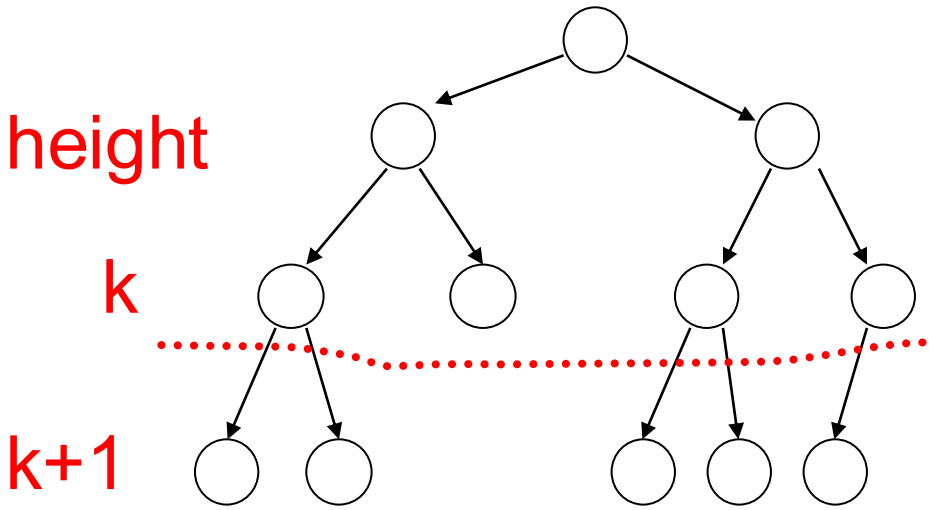
  we need to show that $L(k + 1) <= 2^{k+1}$

Induction Step



From Inductive hypothesis: $L(k) <= 2^k$

Induction Step



$L(k) <= 2^k$

$$L(k+1) <= 2 * L(k) <= 2 * 2^k = 2^{k+1}$$

(we add at most two nodes for every leaf of level k)

Remark

Recursion is another thing

Example of recursive function:

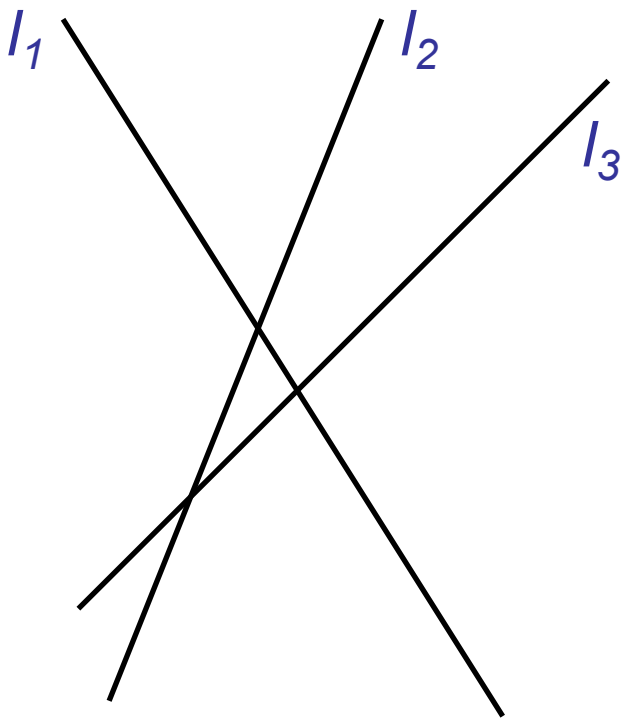
$$f(n) = f(n-1) + f(n-2)$$

$$f(0) = 1, \quad f(1) = 1$$

Example 1.6

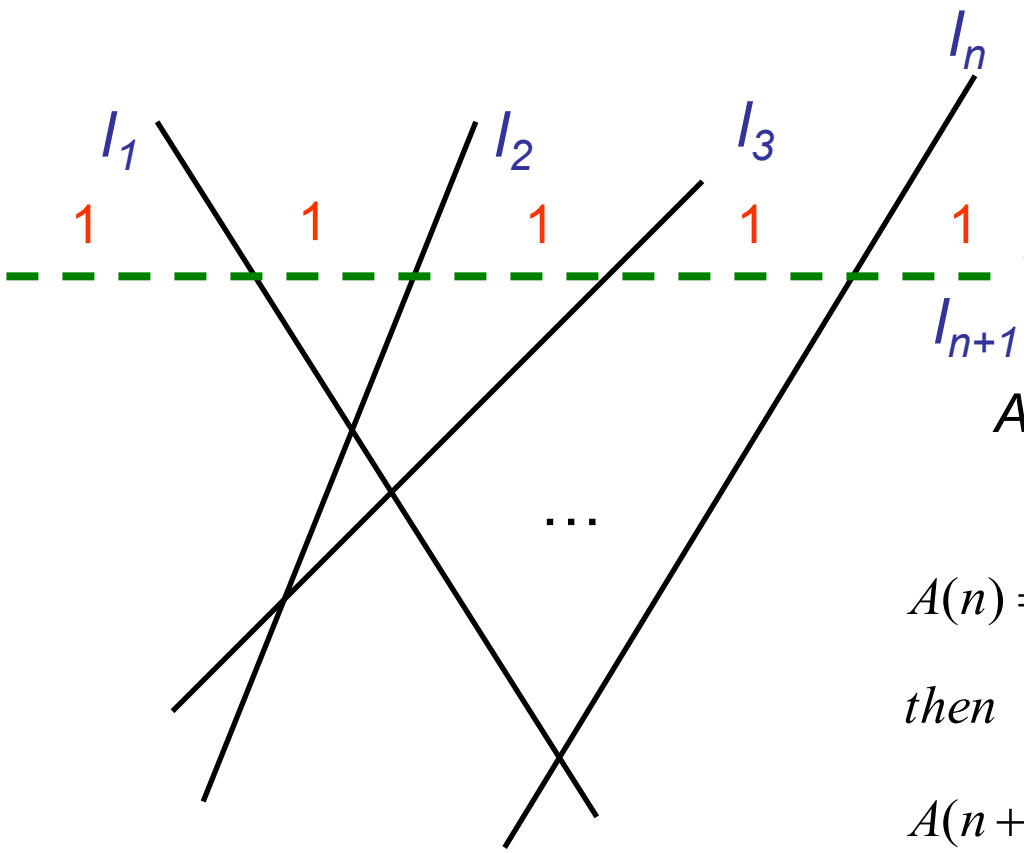
A set $l_1, l_2, \ldots, l_n$ of mutually intersecting straight lines divides the plane into a number of separated regions

1 line → 2 regions, 2 lines → 4 regions, 3 lines → 7 regions



Solve it recursively!!

Example 1.6



Let $A(n)$ denote the number of regions generated by n lines

$A(n+1) = A(n) + n + 1$, n = 1, 2, …,

$A(1) = 2$, $A(2) = 4$, $A(3) = 7$, $A(4) = 11$

$$A(n) = \frac{n(n+1)}{2} + 1$$

then

$$A(n+1) = \frac{n(n+1)}{2} + 1 + n + 1 = \frac{(n+1)(n+2)}{2} + 1$$

Proof by Contradiction

We want to prove that a statement P is true

- we assume that P is false
- then we arrive at an incorrect conclusion
- therefore, statement P must be true

Example 1.7

Theorem: $\sqrt{2}$ is not rational

Proof:

Assume by contradiction that it is rational

$\sqrt{2}$ = n/m

n and m have no common factors

We will show that this is impossible

$\sqrt{2} = n/m \Rightarrow 2m^2 = n^2$

Therefore, n^2 is even $\Rightarrow$ n is even

$n = 2k$

$2m^2 = 4k^2 \Rightarrow m^2 = 2k^2 \Rightarrow$ m is even

$m = 2p$

Thus, m and n have common factor 2

Contradiction!

Outline



Mathematical Preliminaries and Notation



Three Basic Concepts



Some Applications

Three Basic Concepts

- Languages
- Grammars
- Automata (will discuss in Chap. 2)

A language is a set of strings

String: A sequence of symbols from the alphabet

- Examples: “cat”, “dog”, “house”, …
- Defined over an alphabet:

$$\Sigma = \{a, b, c, \ldots, z\}$$

Alphabets and Strings

$$\Sigma = \{a, b\}$$

a

ab $\qquad u = ab$

$abba$ $\qquad v = bbbaaa$

$baba$ $\qquad w = abba$

$aaabbbbaabab$

String Operations

$w = a_1 a_2 \cdots a_n$ $\qquad$ $abba$

$v = b_1 b_2 \cdots b_m$ $\qquad$ $bbbaaa$

Concatenation

$wv = a_1 a_2 \cdots a_n b_1 b_2 \cdots b_m$ $\qquad$ $abbabbbaaa$

$w = a_1 a_2 \cdots a_n$ $ababaaabbb$

Reverse

$w^R = a_n \cdots a_2 a_1$ $bbbaaababa$

String Length

$$w = a_1 a_2 \cdots a_n$$

- Length: $|w| = n$

- Examples: $|abba| = 4$

$|aa| = 2$

$|a| = 1$

Length of Concatenation

$$|uv| = |u| + |v|$$

- Example: $u = aab, \quad |u| = 3$

$$v = abaab, \quad |v| = 5$$

$$|uv| = |aababaaab| = 8$$

$$|uv| = |u| + |v| = 3 + 5 = 8$$

Empty String

- A string with no letters: λ

- Observations: $|\lambda| = 0$

$$\lambda w = w\lambda = w$$

$$\lambda abba = abba\lambda = abba$$

Substring

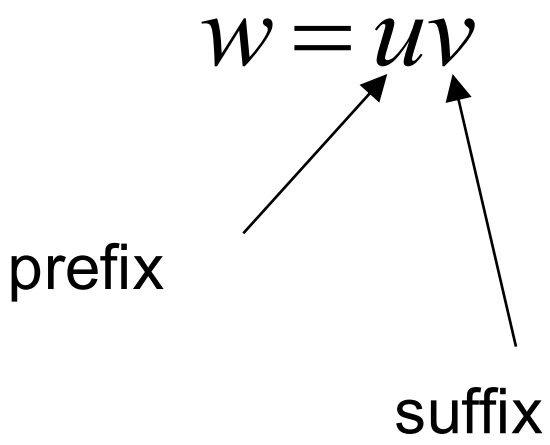
- Substring of string:
  - a subsequence of consecutive characters

String	Substring
$abbab$	ab
$abbab$	$abba$
$abbab$	b
$abbab$	$bbab$

Prefix and Suffix

$abbab$

Prefixes	Suffixes
λ	$abbab$
a	$bbab$
ab	bab
abb	ab
$abba$	b
$abbab$	λ



Another Operation

$$w^n = \underbrace{ww\cdots w}_{n}$$

- Example: $(abba)^2 = abbaabba$

- Definition: $w^0 = \lambda$

$$(abba)^0 = \lambda$$

The * Operation

Σ^* : the set of all possible strings from alphabet Σ

$$\Sigma = \{a, b\}$$

$$\Sigma^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, aab, \ldots\}$$

The + Operation

Σ^+ : the set of all possible strings from alphabet Σ except λ

$$\Sigma = \{a, b\}$$

$$\Sigma^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, aab, \ldots\}$$

$$\Sigma^+ = \Sigma^* - \lambda$$

$$\Sigma^+ = \{a, b, aa, ab, ba, bb, aaa, aab, \ldots\}$$

Languages

A language is any subset of Σ^*

Example: $\Sigma = \{a, b\}$

$$\Sigma^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, \ldots\}$$

Languages: $\{\lambda\}$

(Finite)

$$\{a, aa, aab\}$$

$$\{\lambda, abba, baba, aa, ab, aaaaaa\}$$

Note that:

Sets $\varnothing = \{\} \neq \{\lambda\}$

Set size $|\{\}| = |\varnothing| =$

Set size $|\{\lambda\}| =$

String length $|\lambda| =$

Another Example

- An infinite language $L = \{a^n b^n : n \geq 0\}$

λ

ab

$aabb$

$aaaaabbbbb$

$\in L$

$abb \notin L$

Operations on Languages

- The usual set operations

$$\{a, ab, aaaa\} \bigcup \{bb, ab\} = \{a, ab, bb, aaaa\}$$

$$\{a, ab, aaaa\} \bigcap \{bb, ab\} = \{ab\}$$

$$\{a, ab, aaaa\} - \{bb, ab\} = \{a, aaaa\}$$

- Complement: $\overline{L} = \Sigma^* - L$

$$\overline{\{a, ba\}} = \{\lambda, b, aa, ab, bb, aaa, \ldots\}$$

Reverse

Definition: $L^R = \{w^R : w \in L\}$

Examples: $\{ab, aab, baba\}^R = \{ba, baa, abab\}$

$$L = \{a^n b^n : n \geq 0\}$$

$$L^R = \{b^n a^n : n \geq 0\}$$

Concatenation

Definition: $L_1 L_2 = \{xy : x \in L_1, y \in L_2\}$

Example: $\{a, ab, ba\}\{b, aa\}$

$= \{ab, aaa, abb, abaa, bab, baaa\}$

Another Operation

- Definition: $L^n = \underbrace{LL\cdots L}_{n}$

$$\{a,b\}^3 = \{a,b\}\{a,b\}\{a,b\} =$$

- Special case: $L^0 = \{\lambda\}$

$$\{a, bba, aaa\}^0 = \{\lambda\}$$

More Examples

$$L = \{a^n b^n : n \geq 0\}$$

$$L^2 = \{a^n b^n a^m b^m : n, m \geq 0\}$$

$$aabbaaabbb \in L^2$$

Note that n and m in the above are unrelated

Star-Closure (Kleene *)

- Definition: $L^* = L^0 \bigcup L^1 \bigcup L^2 \cdots$

- Example:

$$\{a,bb\}^* = \left\{ \begin{array}{l} \lambda, \\ a, bb, \\ aa, abb, bba, bbbb, \\ aaa, aabb, abba, abbbb, \ldots \end{array} \right\}$$

Positive Closure

- Definition: $L^+ = L^1 \bigcup L^2 \bigcup \cdots$

$$= L^* - \{\lambda\}$$

$$\{a, bb\}^+ = \left\{ \begin{array}{l} a, bb, \\ aa, abb, bba, bbbb, \\ aaa, aabb, abba, abbbb, \ldots \end{array} \right\}$$

Grammars

Wikipedia says:

- Languages can be described as a system of symbols and the grammars (rules) by which the symbols are manipulated
- Grammar is the study of rules governing the use of language.

Grammars

- Think back to your days of learning English
- Rules for constructing a simple sentence

Sentence = noun phrase + verb phrase

- Noun phrase =
  - Name (Joe)
  - Article + noun (the car)
- Verb Phrase =
  - Verb (runs)
  - Verb + prepositional phrase
- Prepositional Phrase =
  - Preposition + noun phrase (from the car)

Grammars

- Look at the sentence. Is this grammatically correct?

Joe runs from the car.

Sentence = noun phrase + verb phrase
= noun + verb phrase
= Name + verb phrase
= Joe + verb phrase
= Joe + verb + prepositional phrase
= Joe + verb + preposition + noun phrase
= Joe + verb + from + noun phrase
= Joe + verb + from + article +noun
= Joe + verb + from + article +noun
= Joe + verb + from + the + car
= Joe + runs + from + the + car

Valid sentence!

Definition 1.1

- A grammar G is defined as a 4-tuple:

$$G = (V, T, S, P)$$

where

- V is a finite set of variables
- T is a finite set of terminals
- $S \in V$, called start variable
- P is a finite set of production rules

Ex:
G = ({S}, {a, b}, S, P)

P: $S \rightarrow aSb$,
$S \rightarrow \lambda$

Grammars

G = (V, T, S, P)

- Let's formalize this a bit:

  Production rules ($x \rightarrow y$) where $x \in (V, T)^+$, $y \in (V, T)^*$

  They specify how the grammar transforms one string into another

- We say that γ can be derived from α in one step:

  $A \rightarrow \beta$ is a production rule

  $\alpha = \alpha_1 A \alpha_2$

  $\gamma = \alpha_1 \beta \alpha_2$

  $\alpha \Rightarrow \gamma$

- We write $\alpha \overset{*}{\Rightarrow} \gamma$ if γ can be derived from α (or say α derives γ) in zero or more steps.

Definition 1.2

- Let $G = (V, T, S, P)$ be a grammar. Then the set

$$L(G) = \{w \in T^* : S \overset{*}{\Rightarrow} w\}$$

is the language generated by G

- If $w \in L(G)$, then the sequence

$$S \Rightarrow w_1 \Rightarrow w_2 \Rightarrow \ldots \Rightarrow w_n \Rightarrow w$$

is a derivation of the sentence w.

- $S, w_1, w_2, \ldots, w_n$ are called sentential forms

Example 1.11

G = (V, T, S, P)

Consider the grammar:

$G = (\{S\}, \{a, b\}, S, P)$

L(G)?

With P given by

$S \rightarrow aSb,$

$S \rightarrow \lambda$

$L(G) = \{a^n b^n : n \geq 0\}$

Then

$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aabb,$

So we can write

$S \overset{*}{\Rightarrow} aabb$

- String aabb is a sentence in the language generated by G
- aaSbb is a sentential form

Example 1.12

G = (V, T, S, P)

Find a grammar that generates

$$L = \{a^n b^{n+1} : n \geq 0\}$$

Previous example

$G = (\{S\}, \{a, b\}, S, P)$ with P: $S \rightarrow aSb$, $S \rightarrow \lambda$

All we need to do is generate an extra b

$G = (\{S, A\}, \{a, b\}, S, P)$, with productions

$S \rightarrow Ab$,

$A \rightarrow aAb$,

$A \rightarrow \lambda$

Example 1.13

G = (V, T, S, P)

Consider the grammar:

G = ({S}, {a, b}, S, P)

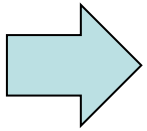
With P given by

$S \rightarrow SS$,

$S \rightarrow \lambda$,

$S \rightarrow aSb$,

$S \rightarrow bSa$,

L(G)? 

Take $\Sigma = \{a, b\}$, and let $n_a(w)$ and $n_b(w)$ denote the number of a's and b's in the string w

$L = \{w: n_a(w) = n_b(w)\}$

Does this grammar indeed generate the language?

Proof by induction!!

Assume that all $w \in L$ with $|w| \leq 2n$ can be derived with G

For n =1, trivial

Example 1.13

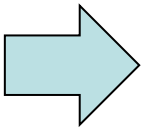
G = (V, T, S, P)

$S \rightarrow SS$,

$S \rightarrow \lambda$,

$S \rightarrow aSb$,

$S \rightarrow bSa$,



Take $\Sigma = \{a, b\}$, and let $n_a(w)$ and $n_b(w)$ denote the number of a's and b's in the string w

$L = \{w: n_a(w) = n_b(w)\}$

Assume that all $w \in L$ with $|w| \leq 2n$ can be derived with G

Take any $w \in L$ of length 2n+2.

If $w = aw_1b$, then w_1 is in L, and $|w_1| = 2n$. By assumption,

$$S \overset{*}{\Rightarrow} w_1$$

Then

$$S \Rightarrow aSb \overset{*}{\Rightarrow} aw_1b = w \quad \text{(so is bSa)}$$

Else

$$S \Rightarrow SS \overset{*}{\Rightarrow} w_1S \overset{*}{\Rightarrow} w_1w_2 = w$$

Equivalent of Grammars

- Two grammars G_1 and G_2 are equivalent if they generate the same language ($L(G_1) = L(G_2)$)

- Example 1.14
- $G_1 = (\{S\}, \{a, b\}, S, P_1)$ with P_1:
  $S \rightarrow aSb$, $S \rightarrow \lambda$
- $G_2 = (\{S, A\}, \{a, b\}, S, P_2)$ with P_2:
  $S \rightarrow aAb| \lambda$, $A \rightarrow aAb| \lambda$

Questions?